Title: A Distributed Multimodal Retrieval System Using CLIP Embeddings

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CSCI 5253 - Datacenter Scale Computing
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Project Overview

- Real-time query handling
- Distributed system focus
- Pretrained model (no training required)

Project Goals

- Build modular system components
- Enable efficient similarity search
- Demonstrate end-to-end pipeline

System Components

- FastAPI
- CLIP
- FAISS
- Redis
- PostgreSQL
- Docker

Architecture Diagram


```
{
  "system_name": "Distributed Multimodal Retrieval System (GeoCLIP)",
  "current_implementation": {
    "api": "FastAPI service handling image uploads, text queries, and debug endpoints",
    "processing_layer": "Coordinates upload/query workflows, simulated queue behavior, caching, and j
ob metrics",
    "queue_layer": "Simulates asynchronous task enqueue/dequeue behavior and worker assignment",
    "database": "PostgreSQL helper layer used for job result tracking",
    "vector_store": "FAISS index used for similarity search",
    "embedding_model": "Pretrained CLIP model used for image and text embeddings"
  },
  "features": [
    "Image-to-text retrieval",
    "Text-to-image retrieval",
    "Simulated asynchronous task queue",
    "Query caching layer",
    "Job tracking and runtime metrics",
    "System health and debug endpoints"
  ],
  "future_distributed_components": [
    "RabbitMQ message broker for true asynchronous job distribution",
    "Dedicated worker services for distributed embedding generation",
    "Expanded PostgreSQL metadata and job tracking",
    "Docker Compose / Kubernetes deployment"
  ]
}
```


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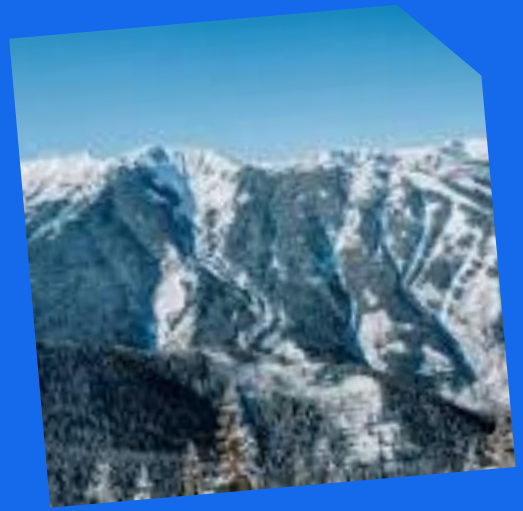
Interactions + Workflows

- API handles requests
- Processing layer coordinates workflow
- CLIP generates embeddings
- FAISS retrieves matches
- Supports image → text and text → image



Debugging & Testing

- Swagger testing
- File handling fixes
- Embedding validation
- End-to-end testing

Performance + Limitations

- Handles small–medium workloads
- Fast FAISS search
- Limited dataset
- No cloud scaling yet

LINK to DEMO

Link to Github REPO